

BAAQAR NAQI

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Summary

Computer Science graduate specializing in ML systems (computer vision, NLP, ensemble modeling) and distributed backend engineering (WebRTC, on-device inference); seeking SDE, ML engineering, or backend engineering roles.

Education

Jaypee University of Engineering and Technology — Guna, MP Aug 2022 – Jun 2026

B.Tech, Computer Science & Engineering GPA: 9.20 / 10.0

Coursework: Machine Learning, Data Structures & Algorithms, Operating Systems, DBMS, AI Applications

Certifications: Machine Learning Specialization — Stanford / Coursera CS50x — Harvard University

City Montessori School — Lucknow, UP 2020 – 2022

ISC Grade 12 (CS elective) — 95.0% ICSE Grade 10 — 98.4% High Distinction, ACER International Benchmark Test (English)

Technical Skills

Languages	Python, C, C++, Java, JavaScript, MySQL
ML / AI	PyTorch, TensorFlow, scikit-learn, OpenCV, ONNX Runtime, Computer Vision, NLP, CNNs, LSTMs, Transformers
Backend & Web	Flask, Django, Node.js, Socket.io, WebRTC, REST APIs
Tools & Systems	Git, Linux, Bash, Docker, Web Workers, AudioWorklet, Cloudflare, Render
Competitive	LeetCode Knight — max rating 1900, 750+ problems solved

Projects

Dyslexia Accessibility ML System — Research Project [Code] [Demo] Python, PyTorch, OpenCV, Flask 2024/2026

- Engineered an end-to-end ML pipeline that detects dyslexic handwriting in raw photographs and generates plain-language reports for educators without specialist equipment or clinical training.
- Built a 3-model ensemble — MLP classifier (EMNIST, ~800K samples), CNN reversal detector (PyTorch + AMP, 91.86% accuracy), and Bidirectional LSTM for sequence anomalies — beating baseline OCR by 25% on 500 real-world samples.
- Devised an OpenCV preprocessing pipeline (adaptive Gaussian thresholding, morphological closing) for non-scanned input; uncovered a critical EMNIST orientation bug (90-degree rotation plus horizontal flip) that produced clean benchmarks while silently breaking real-world inference; containerized with Docker and deployed on Hugging Face Spaces.
- Formulated a sliding-window reversal clustering algorithm calibrated to clinical prevalence rates (Isa et al., 2019), dynamically weighting local peak density against global reversal rate as sample size grows.

PeerSpace — Serverless P2P Collaboration Platform [Code] [Demo] JavaScript, WebRTC, Node.js 2026

- Architected a serverless P2P suite (video, chat, file transfer, shared docs, live transcription) with zero media touching a server; hand-coded the full WebRTC signaling stack (SDP offer/answer, ICE negotiation) in Node.js and Socket.io.
- Structured a full-mesh topology for up to 4 peers across 5 named DataChannels, including a UDP-style unreliable channel for latency-sensitive signals; added chunked binary transfer over a custom framing protocol with sender-side backpressure.
- Added a TURN fallback with credentials from an async REST endpoint, preserving DTLS/SRTP through symmetric NATs.
- Deployed Moonshine ONNX (speech-to-text, 5-15x faster than Whisper in-browser) and DistilBART (meeting summarization) in Web Workers, ensuring no user data leaves the browser; resolved a WebRTC ICE race condition by buffering candidates until setRemoteDescription() completed.

Experience

artifact-to-pwa — Open Source Maintainer [npm] [Code] 2026

- Authored a CLI that converts Claude AI artifacts (JSX/HTML) into cross-platform native desktop apps via a 3-stage pipeline (esbuild bundle to Neutralino runtime injection to binary assembly); implemented SHA-256 integrity verification, a self-healing binary cache, and a window.storage shim — producing ~5 MB installers vs. ~350 MB with Electron.
- Engineered 3 major iterations (PWA to Electron to Neutralino, 70x size reduction), resolving production bugs including cache mismatches, esbuild entry-path failures, and Linux binary permissions; shipped CI/CD via GitHub Actions; 1,372 npm downloads in 3 months with month-over-month growth (305, 420, 604).

Honda Cars India Ltd. — Summer Intern Jun 2025 – Jul 2025

- Developed a Python automation script to consolidate recurring departmental data reports, eliminating ~2 hrs/week of manual effort; presented the design and time-saving projections to stakeholders.
- Audited the department's file-system architecture and delivered a restructuring proposal using the FURPS+ framework; ranked 1st among 6 interns on solution clarity and feasibility.

Leadership

Deputy General Secretary — Development & Innovation Center, JUET Aug 2025 – Jun 2026

- Directed two technical clubs (RoSPinoT — ML & robotics; VR/AR/MR) for 50+ students, running hackathons and mentoring project leads across both clubs.